

Architectural Style Justification

Problem Statement #48 — Incident Escalation & On-Call Rotation Engine

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We chose Microservices Architecture for the Incident Escalation & On-Call Rotation Engine System.

Architectural Choice

The system is decomposed into independently deployable services — Alert Ingestion, Escalation Engine, On-Call Rotation, Notification Dispatch, Incident Data Store, Incident API Gateway, and Post-Mortem Reporting — communicating through well-defined provided/required interfaces, as opposed to a single Layered application or a centralized Client-Server model.

Two Scenario-Related Reasons

1. Independent, workload-driven scaling. During a P1 incident storm, the Notification Dispatch Service must fire push/SMS/email/voice alerts to many engineers within the 3-second budget mandated by NFR-001, while the Post-Mortem Reporting Service and On-Call Rotation Service see comparatively light, steady load. A microservices style lets each service be scaled horizontally on its own, so compute is spent where the spike actually occurs instead of over-provisioning the entire system, which a Layered monolith would force.

2. Fault isolation for a mission-critical escalation path. NFR-002 requires 99.9% uptime specifically for the alert ingestion and escalation pipeline, with failover inside 30 seconds. Because each service is a separate deployable unit, a failure or slow dependency in a non-critical service (e.g., Post-Mortem Reporting) cannot block or crash the Escalation Engine's ability to notify and escalate — a fault stays contained to its own service instance, unlike a single Client-Server node whose failure is a full outage.

Security Advantage

Each service exposes its provided interfaces behind its own authenticated, TLS 1.2+ encrypted boundary (per NFR-001), so service-to-service calls use scoped credentials/mTLS rather than one shared trust domain. If credentials for a single service (e.g., Notification Dispatch) were ever compromised, the blast radius is limited to that service's interfaces rather than exposing the Incident Data Store or the rotation schedule directly, as would be the case in a monolithic Layered design sharing one process boundary.

Performance Benefit

Because the Notification Dispatch Service is a standalone, independently replicable component, additional instances can be deployed behind a load balancer purely to guarantee the sub-3-second dispatch latency required by NFR-001 during peak alert volume, without needing to scale (and pay for) the entire application stack as a Client-Server or Layered architecture would require.